MIDTERM MOCK EXAM – InSy3122 Multimedia Information Systems

Total Points: 30
Time: 60 minutes

Part I: True or False (10 points – 1 point each)

*Write **True** if the statement is correct, **False** otherwise.*

1. Multimedia systems must be computer-controlled and all information must be represented digitally.
2. MIDI files store actual recorded audio, making them larger than MP3 files.
3. Vector graphics lose quality when scaled up, while bitmap images can be scaled to any size without distortion.
4. In an additive color model like RGB, combining equal amounts of red, green, and blue light produces black.
5. Interlaced scanning writes all lines of a frame sequentially, which is the standard for computer monitors.
6. GIF images support up to 24-bit color and are ideal for high-quality photographs.
7. Hypermedia is an extension of hypertext that links multiple media types such as text, audio, and video.

8. Quantization is the process of rounding sampled audio values to the nearest discrete level.
9. The YUV color model separates luminance (brightness) from chrominance (color) to reduce bandwidth and ensure compatibility with black-and-white TVs.
10. WAV is a compressed audio format commonly used for streaming over the Internet.

Part II: Multiple Choice (10 points – 1 point each)

Choose the best answer.

1. Which of the following is NOT a characteristic of a multimedia system?
   a) Computer-controlled
   b) Digitally represented information
   c) Uses only analog storage
   d) Interactive interface
2. Which audio format is a set of instructions rather than a recorded sound?
   a) MP3
   b) WAV
   c) MIDI
   d) AIFF
3. What is the main difference between video and animation?
   a) Video uses sound; animation does not.

b) Video is typically a live recording of moving images; animation is created from static images.
c) Video requires higher resolution than animation.
d) Animation always uses vector graphics.

4. Which image format supports animation and uses a maximum of 256 colors?
   a) JPEG
   b) PNG
   c) TIFF
   d) GIF
5. In the RGB color model, which colors are added to create white?
   a) Red, Yellow, Blue
   b) Red, Green, Blue
   c) Cyan, Magenta, Yellow
   d) Red, Cyan, Green
6. Which of the following is an advantage of vector graphics over bitmap?
   a) Better for photographic images
   b) Smaller file size for complex photographs
   c) Scalability without loss of quality
   d) Supports more color depth
7. What does “sampling rate” refer to in digital audio?
   a) The number of bits used per sample
   b) How often the analog signal is measured per second
   c) The quality of the sound card
   d) The compression ratio of the audio file

8. Which video scanning method writes every other line in one pass and the remaining lines in the next pass?
   a) Progressive scan
   b) Interlaced scan
   c) Linear scan
   d) Raster scan
9. Which color model is subtractive and used primarily for printing?
   a) RGB
   b) YUV
   c) CMYK
   d) YIQ
10. What is the purpose of compression in multimedia systems?
   a) To increase the quality of the media
   b) To reduce the storage space and transmission time
   c) To convert analog to digital signals
   d) To synchronize audio and video

Part III: Short Answer (10 points – 2 points each)

Provide a brief but complete answer.

1. List the four characteristics of a multimedia system.

2. Explain the difference between **linear** and **non-linear** multimedia. Give one example of each.
3. What is the difference between **hypertext** and **hypermedia**?
4. Describe the process of digitizing analog audio, mentioning **sampling** and **quantization**.
5. Why is the YUV color model often used in television and video instead of RGB?

ANSWER KEY

Part I: True or False

1. **True**
2. **False** – MIDI stores instructions, not recorded audio; it is much smaller than MP3.
3. **False** – Vector graphics scale without loss; bitmaps lose quality when enlarged.
4. **False** – In additive color, combining red, green, blue light produces white.
5. **False** – Interlaced scanning writes fields (alternating lines); progressive scanning writes all lines sequentially.
6. **False** – GIF is 8-bit (256 colors); JPEG is better for photographs.

7. **True**
8. **True**
9. **True**
10. **False** – WAV is a large, uncompressed discrete audio format; streaming formats include RealAudio, WMA, etc.

Part II: Multiple Choice

1. c) Uses only analog storage
2. c) MIDI
3. b) Video is typically a live recording of moving images; animation is created from static images.
4. d) GIF
5. b) Red, Green, Blue
6. c) Scalability without loss of quality
7. b) How often the analog signal is measured per second
8. b) Interlaced scan
9. c) CMYK
10. b) To reduce the storage space and transmission time

Part III: Short Answer

1. **The four characteristics are:**
   - Computer-controlled
   - Integrated (all media and devices controlled by a single computer)
   - Information represented digitally

- Interactive interface (user can control/modify the presentation)

2. **Linear multimedia** is presented in a fixed, sequential order with little or no user interaction (e.g., a movie or PowerPoint slideshow).
   Non-linear multimedia allows the user to navigate freely and interact with the content (e.g., a website or video game).
3. **Hypertext** is text that contains links to other text documents.
   Hypermedia extends this concept by linking various media types such as text, graphics, audio, and video, allowing users to navigate through multimedia content.
4. **Digitizing analog audio** involves two main steps:

- **Sampling**: measuring the amplitude of the analog signal at regular time intervals (sampling rate).
- **Quantization**: rounding each sampled value to the nearest discrete level (bit depth) so that it can be represented as a binary number.

5. **YUV is used in television/video** because it separates luminance (brightness) from chrominance (color). The human eye is more sensitive to brightness than color, so the color channels can be compressed or transmitted at lower resolution without noticeable quality loss. Additionally, the Y channel alone provides a compatible signal for black-and-white television sets.